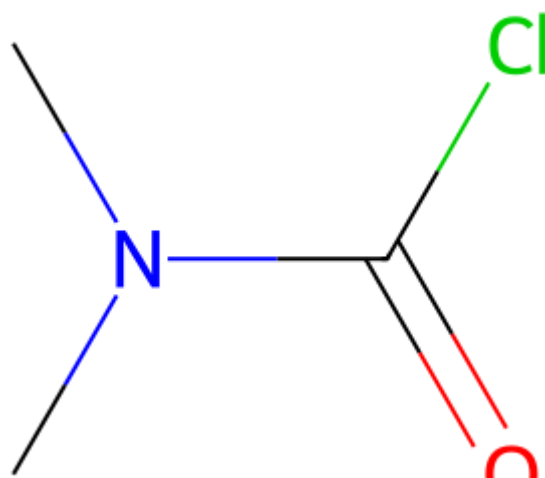




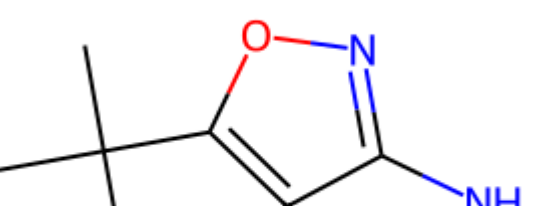
{'analysis_time': 0.0,
'first_solution_time': 0.177804,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 12.097397628904517,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 12.097397628904517,
'retro_star_value': 12.097397628904517}



Reaction
{'analysis_time': 0.177804,
'first_solution_time': 1.097446,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 12.097397628904517,
'retro_star_reaction_number': 12.097397628904517,
'retro_star_rxn_cost': 0.07364654016829848,
'retro_star_value': 12.097397628904517}



{'analysis_time': 0.177804,
'first_solution_time': 1.097446,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 12.02375108873622,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 12.02375108873622,
'retro_star_value': 12.097397628904517}



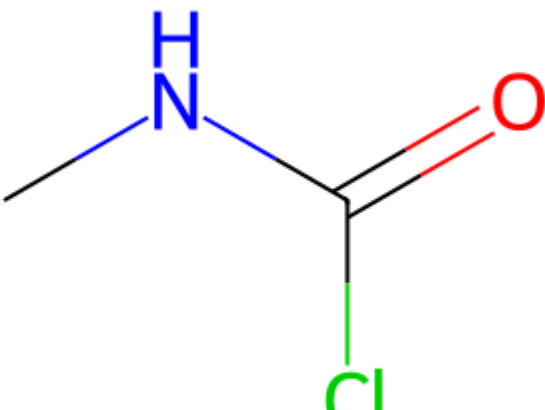
{'analysis_time': 0.177804,
'first_solution_time': 0.177804,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 12.097397628904517}



Reaction
{'analysis_time': 0.285818,
'first_solution_time': 1.097446,
'num_calls_rxn_model': 2,
'num_calls_value_function': 36,
'retro_star_min_cost': 12.02375108873622,
'retro_star_reaction_number': 12.02375108873622,
'retro_star_rxn_cost': 6.214608098422191,
'retro_star_value': 12.097397628904517}



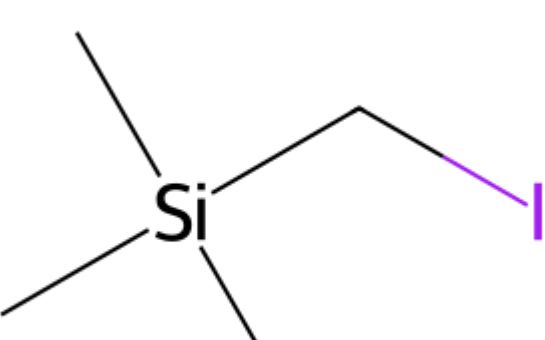
{'analysis_time': 0.285818,
'first_solution_time': 1.097446,
'num_calls_rxn_model': 2,
'num_calls_value_function': 36,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 5.809142990314027,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 5.809142990314027,
'retro_star_value': 12.097397628904517}



{'analysis_time': 0.285818,
'first_solution_time': 0.285818,
'num_calls_rxn_model': 2,
'num_calls_value_function': 36,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 12.097397628904517}



Reaction
{'analysis_time': 1.097446,
'first_solution_time': 1.097446,
'num_calls_rxn_model': 10,
'num_calls_value_function': 297,
'retro_star_min_cost': 23.025850929940457,
'retro_star_reaction_number': 23.025850929940457,
'retro_star_rxn_cost': 23.025850929940457,
'retro_star_value': 29.314105568530948}



{'analysis_time': 1.097446,
'first_solution_time': 1.097446,
'num_calls_rxn_model': 10,
'num_calls_value_function': 297,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 29.314105568530948}